

PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

Software Requirements Specification

Smart Complaint Management System

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06-10-2026



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Revision History

Name	Date	Reason For Changes	Version
Puneeth Yenneti	04-10-2026	Prepared the Introduction, Overall Description.	1.0
Pallavi J	04-10-2026	Prepared the Specific Requirements section, including External Interfaces, and Design Constraints.	1.1
Prajna Holla	04-10-2026	Added the Functional Requirements sections of the SRS.	1.2
Rakshita Muttur	04-10-2026	Added the Non Functional Requirements sections of the SRS.	1.3

Introduction

Purpose

The purpose of a Smart Complaint Management System is to provide a centralized web based system to lodge complaints and monitor their progress, while providing Admins with facilities to review, manage, assign, update and resolve complaints. The system aims to provide a structured mechanism for complaint submission, tracking and resolution.

Intended Audience

This SRS is intended for the following stakeholders:

- **Project Team / Developers:**
  The development team will use this document to understand the functional, non-functional, security, and interface requirements that must be implemented.
- **Testers:**
  The testing team will use the requirements and their unique identifiers to derive test cases and verify that the implemented system satisfies the specified requirements.
- **Project Manager:**
  The project manager will use the SRS to understand the agreed scope, requirements, constraints, and expected system behavior and to track the progress of development.
- **End Users / Complainants:**
  Users can refer to the SRS to understand the intended functionality related to account management, complaint submission, tracking, updates, and notifications.
- **Administrators:**
  Administrators can use the SRS to understand the functionality and privileges provided for complaint management, assignment, resolution, searching, and filtering.
- **Project Evaluators / Faculty:**
  The document provides a structured specification against which the completeness and correctness of the proposed system can be evaluated.

Product Scope

The **Smart Complaint Management System** is a web-based application that provides a centralized platform for submitting, tracking, and managing complaints. Users can create accounts, authenticate, submit complaints, receive unique complaint identifiers, track complaint status and history, and receive notifications about significant updates.

Administrators can view and manage complaints, assign them to appropriate administrators or departments, update complaint information, record resolutions, and search or filter complaints.

The system incorporates role-based access control and security mechanisms to protect user and complaint data. This SRS covers the functional and non-functional requirements of the system and its interactions with users and administrators, but does not specify internal implementation details unless they impose system requirements.

References

[R1] PES University, Department of Computer Science and Engineering, *Software Engineering Mini-Project Deliverables Part-1*, 2026.

[R2] PES University, Department of Computer Science and Engineering, *Software Requirements Specification Template*, Version 1.0.

[R3] IEEE/ISO/IEC, *ISO/IEC/IEEE 29148:2018 — Systems and software engineering — Life cycle processes — Requirements engineering*, 2018.

Overall Description

Product Perspective

The Smart Complaint Management System is a new, self-contained web-based application for managing complaints through a centralized platform. It provides an interface for users to submit and track complaints and an administrative interface for managing, assigning, updating, and resolving complaints.

The system will maintain complaint and user information in a centralized data store and will provide role-based access to system functions. Users and administrators will interact with the system through a web browser.

Product Functions

The major functions of the system are:

- User registration and authentication
- Role-based access control
- Complaint submission and unique complaint ID generation
- Complaint status tracking
- Complaint history and updates
- Administrative complaint management
- Complaint assignment and resolution
- Complaint search and filtering
- Notifications for significant complaint status changes
- Secure management of user and complaint information

User Classes and Characteristics

The system has two primary user classes:

1. User/Complainant:
A user can register and authenticate with the system, submit complaints, view their complaint status and history, provide updates where permitted, and receive notifications regarding their complaints. Users are expected to have basic web browsing knowledge.

2. Administrator:
An administrator has elevated privileges for managing complaints. Administrators can view complaints, assign them to appropriate administrators or departments, update complaint information, record resolutions, and search or filter complaints. Administrators are expected to have basic knowledge of complaint management and the administrative functions provided by the system.

Operating Environment

The Smart Complaint Management System will operate as a web-based application accessible through standard web browsers. The application will be developed using the **Django web framework with Python** and will use **MySQL** for persistent storage of user, complaint, and related system data. Users and administrators will access the system through devices connected to the network using a supported web browser.

Design and Implementation Constraints

The system shall be implemented as a web-based application using **Python and the Django web framework**, with **MySQL** as the database management system. The system shall use a relational database for persistent storage of user and complaint information.

The implementation shall enforce role-based access so that administrative functions cannot be accessed by unauthorized users. User passwords shall not be stored in plaintext and shall use a secure one-way hashing mechanism.

The implementation shall satisfy the functional and non-functional requirements specified in this SRS, including the defined performance, reliability, usability, and security requirements.

2.6 Assumptions and Dependencies

The following assumptions and dependencies apply to the system:

- Users and administrators have access to a supported web browser and network connection.
- Users provide valid information when registering and submitting complaints.
- Administrators are authorized to manage complaints and have the required privileges.
- The system depends on the availability of the Django application server and MySQL database for normal operation.
- Notification functionality depends on the notification mechanism implemented by the system.
- The system depends on the availability and proper configuration of the required Python, Django, MySQL, and operating-system components.

External Interface Requirements

User Interfaces

The Smart Complaint Management System shall provide a web-based user interface for two actors: Complainant/User and Administrator.

The Complainant/User interface shall provide registration and login forms, a complaint submission form, a complaint tracking page, a complaint history view and access to complaint updates and resolution details. The complaint submission form shall allow users to enter the required complaint information, including title, description and category.

The Administrator interface shall provide login access, a complaint management view, complaint details, complaint assignment controls, status update controls and facilities to record resolution details. Administrators shall be able to search and filter complaints using the supported search criteria.

The system shall validate user inputs and display appropriate success messages, validation errors and failure notifications. Access to administrative functions shall be restricted to authorized administrators. The frontend shall be developed using HTML, CSS and JavaScript. Detailed screen layouts and visual design shall be finalized during implementation.

Software Interfaces

The Smart Complaint Management System shall use a MySQL database to store and retrieve user account information, complaint records, complaint status history, assignment information and resolution details.

The frontend, developed using HTML, CSS and JavaScript, shall communicate with the backend application to submit user requests and display system responses. The backend shall process user registration, authentication, complaint submission, complaint tracking, complaint assignment, status updates and complaint resolution operations.

The backend shall validate incoming data, enforce role-based access permissions and perform the required database operations. Each complaint shall be associated with the user who submitted it and shall have a unique complaint identifier.

The backend framework shall be Python with Django Framework. And the database shall be MySQL.

Communications Interfaces

The Smart Complaint Management System shall support communication between the user's web browser and the backend application for exchanging registration details, authentication requests, complaint information, status updates, assignment information and resolution details.

The browser and backend shall communicate using HTTP or HTTPS. HTTPS shall be used in the deployed environment to protect information transmitted between the client and server.

The frontend shall send requests to the backend using the request formats defined during implementation. The backend shall return responses indicating successful operations, validation failures, authentication failures or unauthorized access attempts.

The system shall validate incoming requests and enforce access permissions before processing protected operations. Passwords and other sensitive information shall not be exposed in application responses. Communication failures shall be handled appropriately, and unsuccessful operations shall not be reported as successful.

The API endpoints, request and response formats, session-management mechanism and deployment configuration shall be finalized during software design. Email or other external notification services are not included in the current scope unless approved by the project team.

Analysis Models

The analysis models for the Smart Complaint Management System shall include a UML use-case diagram representing the interactions between the Complainant/User, Administrator and the system.

The use-case diagram shall represent the major functions of the application, including user registration, authentication, complaint submission, complaint tracking, viewing complaint history, providing additional information, complaint assignment, status management, complaint search and filtering, and complaint resolution.

An entity-relationship diagram shall represent the relationships between user accounts and complaint records, including relevant status history, assignment and resolution information. The detailed database structure shall be finalized during the database-design phase.

Functional Requirements

System Features

=> User Account Management

Description and Priority

This feature allows new users to register for an account and registered users to authenticate themselves before accessing protected functions. The system shall distinguish between Complainant/User and Administrator roles and provide access privileges according to the assigned roles.

Priority: High.

Stimulus/Response Sequences

1. A new user opens the registration page and enters the required registration details.
2. The system validates the submitted information and creates the account if the details are valid.
3. A registered user enters their credentials on the login page.
4. The system validates the credentials and grants access if authentication is successful.
5. The system provides access to functions permitted for the authenticated user's role.
6. If the registration details are invalid or the credentials are incorrect, the system displays an appropriate error message.
7. If a user attempts to access an unauthorized function, the system denies access.

Functional Requirements

FR-01 — User Registration

The system shall allow a new user to create an account by providing the required registration details. The system shall validate the submitted information and reject incomplete or invalid registration requests.

FR-02 — User Authentication

The system shall authenticate registered users using their registered credentials before granting access to protected functions.

FR-03 — Role-Based Access

The system shall provide different access privileges for Complainants/Users and Administrators based on their assigned roles.

Complaint Submission

Description and Priority

This feature allows authenticated users to lodge complaints by providing the required complaint information. The system shall validate the submitted details, generate a unique complaint identifier and display confirmation after successful submission.

Priority: High.

Stimulus/Response Sequences

1. An authenticated user opens the complaint submission form.
2. The user enters the complaint title, description, category and other required supporting information.
3. The user submits the complaint.
4. The system validates the submitted information.
5. If the information is valid, the system saves the complaint and generates a unique complaint identifier.
6. The system displays the complaint identifier and submission confirmation to the user.
7. If mandatory information is missing or invalid, the system displays an appropriate error message and requests correction.

Complaint Submission

FR-04 — Complaint Submission and ID Generation

The system shall allow an authenticated user to submit a complaint by providing the required complaint details. Upon successful submission, the system shall generate a unique complaint identifier and display a submission confirmation to the user.

Complaint Tracking

This feature allows users to monitor the progress of their submitted complaints and view their status history and relevant updates. It also supports the addition of information and updates according to the permissions assigned to users and administrators.

Priority: High.

Stimulus/Response Sequences

1. A user logs in and opens the complaint tracking section.
2. The system displays the complaints submitted by that user and their current statuses.
3. The user selects a complaint to view its details, status history and available updates.
4. An authorized user or administrator adds an update or additional information according to their assigned privileges.
5. The system validates and records the update against the relevant complaint.
6. The system displays the updated complaint information.
7. If the user attempts to access a complaint without the required permissions, the system denies access.

The proposed complaint statuses are Submitted, Under Review, Assigned, In Progress and Resolved. Rejected and Closed may be included if approved by the project team.

FR-05 — View Complaint Status

The system shall allow a user to view the current status of complaints submitted by that user.

FR-06 — Complaint History

The system shall maintain the status history and relevant updates associated with each complaint.

FR-07 — Complaint Update

The system shall allow authorized users or administrators to add updates to a complaint according to their assigned privileges.

Administrative Management

Description and Priority

This feature allows authorized administrators to view, manage and assign submitted complaints, update their statuses and record resolution details. It

supports the administrative processing of complaints from submission through resolution.

Priority: High.

Stimulus/Response Sequence

1. An administrator logs in using valid credentials
2. The system provides access to the administrative complaint management interface.
3. The administrator views the submitted complaints and selects a complaint.
4. The administrator reviews the complaint details and assigns it to an appropriate administrator or responsible department.
5. The administrator updates the complaint status as processing progresses.
6. When the complaint is resolved, the administrator records the resolution details and marks the complaint as resolved.
7. The system saves the changes and displays the updated complaint information.
8. If an unauthorized user attempts to perform an administrative operation, the system denies the request.

FR-08 — Admin Complaint Management

The system shall allow an administrator to view and manage complaints submitted by users.

FR-09 — Complaint Assignment

The system shall allow an administrator to assign a complaint to an appropriate administrator or responsible department.

FR-10 — Complaint Resolution

The system shall allow an authorized administrator to update a complaint with its resolution and mark the complaint as resolved.

Search and Monitoring

Description and Priority

This feature allows administrators to locate and monitor complaints using relevant complaint attributes. It supports efficient retrieval of complaint records for administrative review and processing.

Priority: Medium.

Stimulus/Response Sequences

1. An authorized administrator opens the complaint management interface.
2. The administrator enters a complaint identifier or selects one or more available filtering criteria.
3. The system searches the complaint records using the specified criteria.
4. The system displays the complaints matching the search or filter conditions.
5. If no matching complaints are found, the system displays an appropriate message.

FR-11 — Complaint Search and Filtering

The system shall allow administrators to search and filter complaints based on attributes such as complaint ID, category, status and submission date.

FR-12 — Complaint Notifications

The system shall notify the relevant user when significant changes occur to the status of their complaint.

Nonfunctional Requirements

Performance

NFR-01 — Response Time
The system shall respond to normal user requests within **3 seconds** under the specified normal operating load.

NFR-02 — Concurrent Users
The system shall support at least **100 concurrent authenticated users** without failure under the defined test environment.

Availability / Reliability

NFR-03 — Data Persistence
The system shall persist successfully submitted complaints so that complaint information is not lost during normal application operation.

NFR-04 — Error Recovery
The system shall display an appropriate error message when an operation fails and shall not create duplicate complaint records as a result of a failed submission.

Usability

NFR-05 — Usability
The system shall allow a registered user to submit a complaint through the web interface without requiring technical knowledge of the underlying system.

NFR-06 — Validation Feedback
The system shall display clear validation messages when required complaint fields are missing or contain invalid data.

Security

Security Objectives

SO-01 — Authentication and Access Control
Protect system functionality and complaint information from unauthorized access.

SO-02 — Data Protection
Protect user credentials and complaint information against unauthorized disclosure or modification.

Security Requirements

NFR-07 — Authentication Security
The system shall require authentication before allowing users to access protected complaint information or administrative functions.

NFR-08 — Authorization
The system shall restrict administrative functions to authenticated users with administrator privileges.

NFR-09 — Password Protection
The system shall store user passwords using a secure one-way password hashing mechanism rather than storing plaintext passwords.

NFR-10 — Session Security
The system shall invalidate an authenticated session after logout.

Business Rules

1. A user must be authenticated before submitting a complaint or accessing protected complaint information.
2. Each successfully submitted complaint shall have a unique complaint identifier.
3. Each complaint shall be associated with the user who submitted it.
4. A Complainant/User shall be permitted to view their own complaints but shall not have access to administrative functions.
5. Only an authorized Administrator shall be permitted to assign complaints, manage complaint statuses and record resolution details.
6. Complaint statuses shall follow the status values and transitions defined for the application. The proposed statuses are Submitted, Under Review, Assigned, In Progress and Resolved. Rejected and Closed shall be included only if approved by the project team.
7. A complaint shall be marked as Resolved only after the authorized Administrator has recorded its resolution details.
8. Complaint updates shall be permitted only to users authorized to perform the corresponding operation.

Other Requirements

Database Requirements

1. The system shall use MySQL as its database management system.
2. The database shall store user account information, complaint records, complaint status history, assignment information and resolution details as required by the approved system features.
3. Each complaint shall have a unique identifier.
4. Relationships between users and complaints shall preserve the association between each complaint and its submitting user.
5. The database shall enforce appropriate constraints to maintain data consistency and referential integrity.
6. Database access shall be performed through the backend application, and database credentials shall not be exposed to the client-side interface.

Technology Requirements

1. The frontend shall be developed using HTML, CSS and JavaScript.
2. The backend shall be developed using Python with Django Framework.
3. The system shall use MySQL for persistent data storage.

Documentation and Testing Requirements

1. The system shall be documented through the Software Requirements Specification, Software Architecture and Design Specification, and Software Test Plan.
2. The implemented system shall be tested against the approved functional and nonfunctional requirements.
3. The final implementation shall maintain consistency between the approved requirements, software design and test cases.

Appendix A: Glossary

Administrator: An authorized user responsible for managing complaints, assigning complaints, updating complaint statuses and recording resolution details.

Authentication: The process of verifying a user's identity using registered credentials before granting access to protected functions.

Complainant/User: A registered user who submits complaints and tracks their progress through the Smart Complaint Management System.

Complaint: A record submitted by a user describing an issue that requires review and resolution.

Complaint Category: A classification used to organize complaints according to their type.

Complaint ID (Complaint Identifier): A unique identifier assigned to each successfully submitted complaint.

Complaint History: A record of status changes and relevant updates associated with a complaint.

Complaint Status: The current stage of a complaint in the complaint management process, such as Submitted, Under Review, Assigned, In Progress or Resolved.

Complaint Resolution: The outcome or corrective action recorded by an authorized Administrator to address a complaint.

Database: An organized collection of data used to store and retrieve user accounts, complaint records and related information.

DBMS (Database Management System): Software used to create, manage and access databases. MySQL is the selected DBMS for this project.

FR (Functional Requirement): A specification describing a function or behavior that the system must provide.

Frontend: The user-facing part of the application through which users interact with the system.

HTTPS (Hypertext Transfer Protocol Secure): A communication protocol that protects data transmitted between a web browser and a server using encryption.

MySQL: The relational database management system selected to store and manage the application's data.

RBAC (Role-Based Access Control): An access-control approach in which permissions are granted according to a user's assigned role.

SAD (Software Architecture and Design Specification): A document describing the system's architecture, components, interfaces and design decisions.

SRS (Software Requirements Specification): A document describing the system's scope, functional requirements, nonfunctional requirements, interfaces and constraints.

Status Update: Information recorded to reflect a change in a complaint's status or progress.

UI (User Interface): The screens, forms and controls through which users interact with the application.

UML (Unified Modeling Language): A standardized modeling language used to represent software structure and behavior, including use-case diagrams.

Validation: The process of checking submitted information to ensure that it satisfies the system's defined requirements.

Web Browser: Software used to access and interact with the web-based Smart Complaint Management System.

Appendix B: Field Layouts

An Excel sheet containing field layouts, field properties/attributes and report requirements for the Smart Complaint Management System.

Sample Sheet: Information Required to Register a User

Field	Length	Datatype	Description	Is Mandatory
User ID	20	Alphanumeric	Unique identifier for the registered user	Yes
Full name	100	String	Full name of the user	Yes
Email Address	254	Alphanumeric	Email address used for account identification	Yes

Password	128	String	Password provided during registration, stored securely as a password hash	Yes
Role	20	Alphanumeric	Role assigned to the account : user or administrator	Yes

Sample Sheet: Information Required to Submit a Complaint

Field	Length	Datatype	Description	Is Mandatory
Complaint ID	20	Alphanumeric	Unique Identifier generated for each complaint	Yes
User ID	20	Alphanumeric	Identifier of the user who submitted the complaint	Yes
Complaint Title	150	String	Short title describing the complaint	Yes
Complaint Description	-	Text	Detailed description of the issue	Yes
Complaint Category	50	String	Category selected for the complaint	Yes

Supporting Information	-	Text	Additional information relevant to the complaint	TBD
Submission Date & Time	-	DateTime	Date and time when the complaint is submitted successfully	Yes
Complaint Status	30	Alphanumeric	Current status of the complaint	Yes
Assigned Administrator / Department	100	String	Administrator or department responsible for handling the complaint	TBD
Resolution Details	-	Text	Details of the resolution recorded by an authorized Administrator	TBD

Sample Sheet: Complaint Status History

Field	Length	Datatype	Description	Is Mandatory
History ID	20	Alphanumeric	Unique Identifier for a status-history entry	Yes
Complaint ID	20	Alphanumeric	Identifier of the associated	Yes

			complaint	
Previous Status	30	Alphanumeric	Status before the recorded change, if applicable	No
New Status	30	Alphanumeric	Status after the recorded change	Yes
Updated By	20	Alphanumeric	Identifier of the user or administrator who made the update	Yes
Update Details	-	Text	Description of the update, where applicable	No
Update Date&Time	-	DateTime	Date and time when the time was recorded	Yes

Sample Report Requirements

1. User Complaint Report

The report shall include the following fields:

- Complaint ID
- Complaint Title
- Complaint Category
- Submission Date and Time
- Current Complaint Status
- Assigned Administrator/Department, if applicable
- Resolution Details, if available

Access: A Complainant/User shall be able to view complaints submitted by that user. An Administrator shall be able to view complaints permitted by the administrator's access privileges.

2. Administrative Complaint Report

The report shall include the following fields:

- Complaint ID
- Complainant/User ID
- Complaint Title
- Complaint Category
- Submission Date and Time
- Current Complaint Status
- Assigned Administrator/Department, if applicable
- Latest Update Date and Time
- Resolution Details, if available

Access: Authorized Administrators shall be able to view and filter complaint records based on the supported search criteria, including complaint ID, category, status and submission date.

3. Complaint History Report

The report shall include the following fields:

- Complaint ID
- Previous Status, if applicable
- New Status
- Updated By
- Update Details, if applicable
- Update Date and Time

Access: The report shall be available only to users authorized to view the corresponding complaint history.

Appendix C: Requirement Traceability Matrix

Sl. No	Requirement ID	Brief Description of Requirement	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
1	FR-01	User Registration	SAD Sec. 3.4: Auth & Session	SAD Sec. 4.5: UX Design	-	TC-FR-01	ST-FR-01

			Service, Web UI				
2	FR-02	User Authentication	SAD Sec. 3.4: Auth & Session Service	SAD Sec. 3.9: Security Architecture	-	TC-FR-02	ST-FR-02
3	FR-03	Role-Based Access	SAD Sec. 3.4: Auth Service, Controllers	SAD Sec. 3.9: RBAC Enforcement	-	TC-FR-03	ST-FR-03
4	FR-04	Lodge Complaint	SAD Sec. 3.4: Complaint Service, Repository / DAO	SAD Sec. 4.2: Flow 1	-	TC-FR-04	ST-FR-04
5	FR-05	ComplaintID Generation	SAD Sec. 3.4: Complaint Service	SAD Sec. 4.1: Status Workflow; Sec. 4.5: UX Design	-	TC-FR-05	ST-FR-05
6	FR-06	View Complaint Status	SAD Sec. 3.4: Complaint Service, Complaint_ History	SAD Sec. 4.1: Status History	-	TC-FR-06	ST-FR-06
7	FR-07	Complaint History	SAD Sec. 3.4: Complaint Service, Admin Management Service	SAD Sec. 4.2: Flow 2	-	TC-FR-07	ST-FR-07

8	FR-08	Complaint Update	SAD Sec. 3.4: Admin Management Service	SAD Sec. 4.5: Admin Dashboard	-	TC-FR-08	ST-FR-08
9	FR-09	Admin Complaint Management	SAD Sec. 3.4: Admin Management Service	SAD Sec. 4.2: Flow 2	-	TC-FR-09	ST-FR-09
10	FR-10	Complaint Assignment	SAD Sec. 3.4: Admin Management Service	SAD Sec. 4.1: Status Workflow	-	TC-FR-10	ST-FR-10
11	FR-11	Complaint Resolution	SAD Sec. 3.4: Admin Management Service	SAD Sec. 4.2: Flow 2	-	TC-FR-11	ST-FR-11
12	FR-12	Complaint Search & Filtering	SAD Sec. 3.4: Search \& Filter Module	SAD Sec. 4.5: Admin Dashboard	-	TC-FR-12	ST-FR-12